

Shyam Sundar Debsarkar

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AI Researcher

PhD (ABD) researcher in Computer Science at the University of Cincinnati specializing in AI for Computer Vision and Medical Imaging. Experienced in Deep Learning, Large Language Models (LLMs), Vision Language Models (VLMs), Agentic AI, Multimodal Learning, RAG, and Generative AI models. Skilled at building clinically relevant interpretable pipelines integrating imaging, omics, and textual modalities. Over 6 years of combined experience spanning industrial software architecture and AI research. Aspiring AI Scientist passionate about researching and applying cutting-edge AI technologies to solve real world problems.

EDUCATION

University of Cincinnati, USA

Ph.D. (ABD), Computer Science

2026

Technical University of Munich, Germany

M.S., Informatics

West Bengal University of Technology, India

B.Tech., Electronics & Communication Engineering

EXPERIENCE

Doctoral Researcher

University of Cincinnati

2022 - Present

- Led a cross-functional team to develop a multimodal Brain Glioma classification pipeline using TCGA dataset; achieved AUC 0.967 / Accuracy 0.936
- Developed a diffusion-based histopathology generator with point/map conditioning; achieved FID 36.4 and improved downstream segmentation to Dice 0.83 (+~20% vs. baseline)
- Enhanced interpretability in Multi-Agentic LLM arbitration framework solving traditional agents' decision-making problem beating state-of-the-art methods over imaging datasets. Achieving ~15% accuracy improvement
- Evaluated multiple VLM embeddings on LungMAP WSIs (>30k patches) for unsupervised biomarker discovery; ran large-scale experiments on A100/V100 clusters
- Evaluation of different segmentation and detection models (SAM, YOLO) in small particle analysis, e.g. Nuclei

AI Graduate Research Assistant

Procter & Gamble

2025 - Present

- Built a RAG-based lab-specific LLM platform for P&G scientists, integrating vector search over internal scientific documents; achieving 42% efficiency improvement in lab workflows
- Extended the platform into an agentic system using LangGraph and LangChain, enabling multi-step scientific reasoning and tool-use; exposed via FastAPI endpoints consumed by internal lab applications
- Applied DPO-based post-training alignment to the platform's domain LLM on lab-specific preference pairs, improving response quality by ~35% on BLEU and ROUGE-L evaluation
- Separately, built deep learning pipelines for TEM image denoising, segmentation, and continual learning, improving image clarity and analysis accuracy by ~30%

Software Architect

Jambit GMBH, Munich, Germany

2018 - 2022

- Led cross-functional team involving BMW in road-tracker map development with 35% latency reduction
- Delivered solution to Hoffmann utilizing full-stack platform (Java/Spring + React) with 40% faster CI/CD Deployment

Software Developer

2017-2018

Oncare GMBH, Munich, Germany

- Built React Native based mobile app serving >500 clinics with enhanced real-time sync reliability by ~25%

SKILLS

- Deep Learning
- Computer Vision, Medical Image Analysis
- TEM (Transmission Electron Microscope) images
- LLMs / VLMs
- Multimodal AI
- Multi-Agent AI, Agentic AI
- LangGraph, LangChain
- LLM post-training techniques - PPO, DPO, RLHF
- FastAPI
- Generative AI, Diffusion Models, GANs
- Retrieval-Augmented Generation (RAG)
- Python (NumPy, Pandas, Matplotlib, scikit-learn, OpenCV), PyTorch, TensorFlow
- High Performance Computing (HPC) Clusters, A100, V100, H100
- Docker, Git, AWS / Azure
- Website and Mobile App development, Javascript, Java, NodeJS

PUBLICATIONS

- Deep Learning-Based IDH1 Mutation Prediction via WSI + Clinical Fusion. CBM, 2024
- A Multi-Expert Deep Learning Framework with LLM-Guided Arbitration for Multimodal Histopathology Prediction. Computerized Medical Imaging and Graphics (CMIG), 2026
- SpaDiffHis: Diffusion-Based Histopathology Generation with Point-Map Conditioning IEEE JBHI, 2026
- Unsupervised Biomarker Discovery via Vision-Language Model Fusion. Neural Computing & Applications (NCAA), 2025
- Advancements in Automated Nuclei Segmentation using YOLO-Driven Approaches. CBM, 2025
- Multimodal Ensemble Fusion Deep Learning for Glioma Subtype Classification. IEEE Access, 2025
- Vision Transformers and CNN-Based Knowledge Distillation for Histopathology. FSDM, 2024
- Vision Transformers for Histopathological Image Classification with Efficient Head Pruning, KES, 2025
- Ensemble Deep Learning Model for Glioma Histopathological Image Classification, KES, 2025
- Public H&E Imaging Datasets of Pathology for Artificial Intelligence: A Review, Journal of imaging informatics in medicine, 2026
- End-to-End Multimodal Multiple Instance Learning for Cancer Histopathology Classification with Dual-Attention Fusion, JMS, 2026

Under Review

- Optimizing Conformal Prediction Sets for Pathological Image Classification (submitted to Pattern Recognition)
- Multiple Instance Learning using Reduced Deep Learning Model for Glioma Subtype Classification (submitted to IMA, 2025)